Task 3 Tool Design: Jurisdiction-Aware AI Credit Fairness Checker

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1. Plain-Language Design Summary

This document constitutes the required one-page plain-language summary (Section 1) and the senior management pitch (Sections 2-6) for the Task 3 working prototype. The prototype implements Option A (Working Prototype) with additional quantitative analysis elements drawn from Option C.

The tool is a jurisdiction-aware compliance checker for AI-driven credit scoring, designed for a Lendable / Zable-style consumer lending platform operating across the United States and the United Kingdom. It takes synthetic applicant data, trains a logistic regression credit risk model, and evaluates the same model output through two separately configured regulatory rule sets. Sensitive demographic attributes are excluded from model training but retained for post-decision fairness auditing.

The central design claim is that the same model output carries different compliance meanings depending on jurisdiction. In the latest demo run, the US configuration returns 'No major issues detected' while the UK configuration returns 'Review Required'. The US configuration enforces an approval-disparity threshold of 0.80 (four-fifths rule convention from fair lending practice) and requires specific adverse action reasons for rejected applicants. The UK configuration enforces a stricter disparity threshold of 0.90, consistent with the FCA Consumer Duty's outcome-monitoring philosophy, and additionally requires model card documentation, human oversight evidence, and board-level remediation governance.

The tool does not provide legal advice, determine that unlawful discrimination has occurred, or replace human judgement. These boundaries are deliberate: the prototype surfaces regulatory risk and governance gaps for human review. The design explicitly chooses not to produce a compliance certificate or a single pass/fail score because such outputs would serve documentation over genuine risk detection.

2. Assignment Requirement Mapping

The table below maps each Option A and Option C requirement to the specific component in the prototype that satisfies it.

Requirement	Where satisfied
Jurisdiction config layer	src/config_rules.py — US and UK rules, thresholds,
Live/simulated data input	data/synthetic_applicants.csv — 1,500 synthetic
Different output by juris.	outputs/compliance_reports.json — US 'No major issues',
Model card / governance stub	docs/Task3_Model_Card.md — purpose, inputs, exclusions,
Quantitative metric	Approval disparity ratio, approval rate, ROC-AUC,
Threshold methodology	US threshold 0.80 (four-fifths rule); UK threshold 0.90
Sensitivity analysis	outputs/threshold_sensitivity.csv — disparity ratio vs.
Drift simulation	outputs/drift_comparison.csv — recession income shock
Failure modes	False positives, false negatives, drift, jurisdiction
Management pitch	Sections 3-6 below; Limitations in Section 6

3. Why Is This Product Right for the Company?

The hypothetical RegTech company (FairLens RegTech) is positioned as a specialist compliance software provider for mid-market AI-lending platforms expanding across the US and UK. Its core competence is translating regulatory expectations into executable, versioned operational checks — not providing legal advice. This prototype directly reflects that positioning. AI credit scoring creates a recurring, jurisdiction-specific monitoring need: the same model must be re-evaluated every time regulation changes, portfolios shift, or a new market is entered. A configurable, automated tool is the natural product for that need.

The commercial rationale is also sound. US ECOA enforcement settlements have ranged from USD 1 million to over USD 100 million (BancorpSouth 2016, Trident Mortgage 2021), making a USD 100,000-150,000 annual compliance tool a rational insurance purchase. In the UK, the FCA has signalled that board-level evidence of Consumer Duty outcome monitoring is a prerequisite for ongoing authorisation in consumer credit. The product converts both of these compliance obligations into a repeatable software workflow rather than a recurring manual exercise.

4. What Problem Does This Tool Solve That a Memo or Spreadsheet Cannot?

A memo can describe regulatory differences between the US and the UK, but it cannot rerun those checks when a new applicant dataset arrives, when a model threshold is adjusted, or when a regulation is updated. It is a snapshot, not a workflow. A spreadsheet can store individual metrics but does not naturally preserve rule provenance (which version of Regulation B?), model assumptions, adverse action explanation logic, and drift evidence in a single reproducible artefact.

This tool provides three specific capabilities that neither a memo nor a spreadsheet can replicate. First, the jurisdiction configuration layer (src/config_rules.py) stores regulatory parameters as versioned, source-linked records. When the Trump administration's Regulation B rewrite changed the US explainability standard in April 2026, a compliance team using a spreadsheet would need to manually update every downstream document. This tool requires a single rule-version update. Second, the tool generates adverse action explanations for every rejected applicant as part of the same pipeline that checks disparity ratios — the output is consistent and traceable, not manually assembled. Third, the drift simulation and sensitivity analysis produce quantitative evidence for regulators, not narrative assertions. The compliance finding is connected to the data and the rule in one JSON report.

5. What Regulatory Divergence Does the Tool Handle, and How?

The tool handles the divergence between US fair lending obligations (anchored in ECOA and Regulation B) and UK outcome-based governance obligations (anchored in the FCA Consumer Duty, PS22/9).

Within the US, the political landscape has itself shifted. The Biden-era CFPB (October 2023) required specific, model-derived adverse action reasons for AI-driven credit decisions, tightening explainability obligations. The Trump-era CFPB's Regulation B Subpart A rewrite (April 2026) reduced that pressure. The prototype's rule_version and last_reviewed fields in config_rules.py are specifically designed to track these mid-period political changes, allowing compliance teams to operate against the correct version of the rule rather than an outdated snapshot.

The structural divergence between US and UK is operationalised through three configurable differences. (1) Disparity threshold: the US configuration flags approval-rate disparity below 0.80; the UK configuration flags disparity below 0.90, reflecting the Consumer Duty's stricter outcome-monitoring stance. (2) Governance requirements: the UK configuration requires model card documentation, human oversight evidence, and Consumer Duty outcome monitoring flags; the US configuration requires model card documentation (consistent with OCC 2026-13 model lifecycle expectations) but not the UK's board-level remediation governance layer. (3) Adverse action logic: both configurations require specific adverse action reasons, but the UK configuration additionally requires evidence that governance processes are embedded in the product lifecycle — not just that individual denials are explained.

The demo result confirms the jurisdiction logic is real and not decorative. Using the same 1,500-applicant synthetic portfolio and the same logistic regression model at a 0.50 decision threshold, the US configuration returns 'No major issues detected' because all group-level approval disparity ratios exceed 0.80. The UK configuration returns 'Review Required' because the nationality disparity ratio (0.898) and the ethnicity disparity ratio (0.858) both fall below the stricter 0.90 UK threshold.

Figure 1

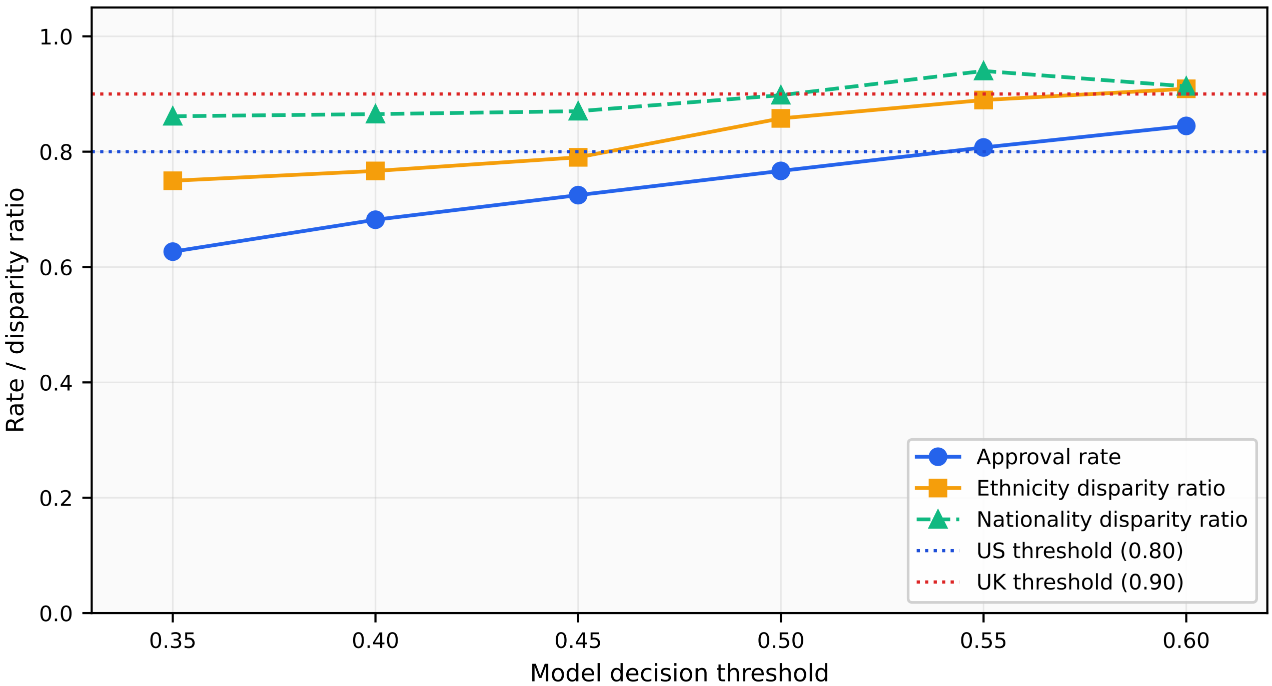

Figure 1: Threshold sensitivity analysis. As the model decision threshold rises, approval rates and group disparity ratios both improve. The ethnicity disparity ratio crosses the US threshold (0.80, blue dotted) at approximately threshold=0.50, and approaches but does not consistently exceed the UK threshold (0.90, red dotted) until threshold=0.60. This demonstrates that the same portfolio clears the US compliance check but fails the UK check across a wide range of model settings — confirming that the jurisdiction divergence is structural, not marginal.

5.1 Drift Simulation

The drift simulation models a recession scenario in which applicant incomes are reduced by 12% and debt-to-income ratios are inflated by 18%. The shock is applied to the identical 1,500-applicant baseline cohort (same random seed), isolating the effect of the economic shock from variation in population composition. Figure 2 shows the effect on approval rate and group-level disparity ratios.

Under the recession shock, the approval rate falls from approximately 0.767 (baseline) to approximately 0.721 — a meaningful reduction in credit access driven by the income and DTI deterioration. More importantly, the ethnicity disparity ratio falls from approximately 0.858 to approximately 0.799, crossing below the US compliance threshold of 0.80. A portfolio that currently clears the US check would therefore fail that same check under recession conditions, using the same fixed model and the same threshold. This result illustrates the core argument for continuous monitoring: a one-time validation is not sufficient to guarantee ongoing compliance.

Figure 2

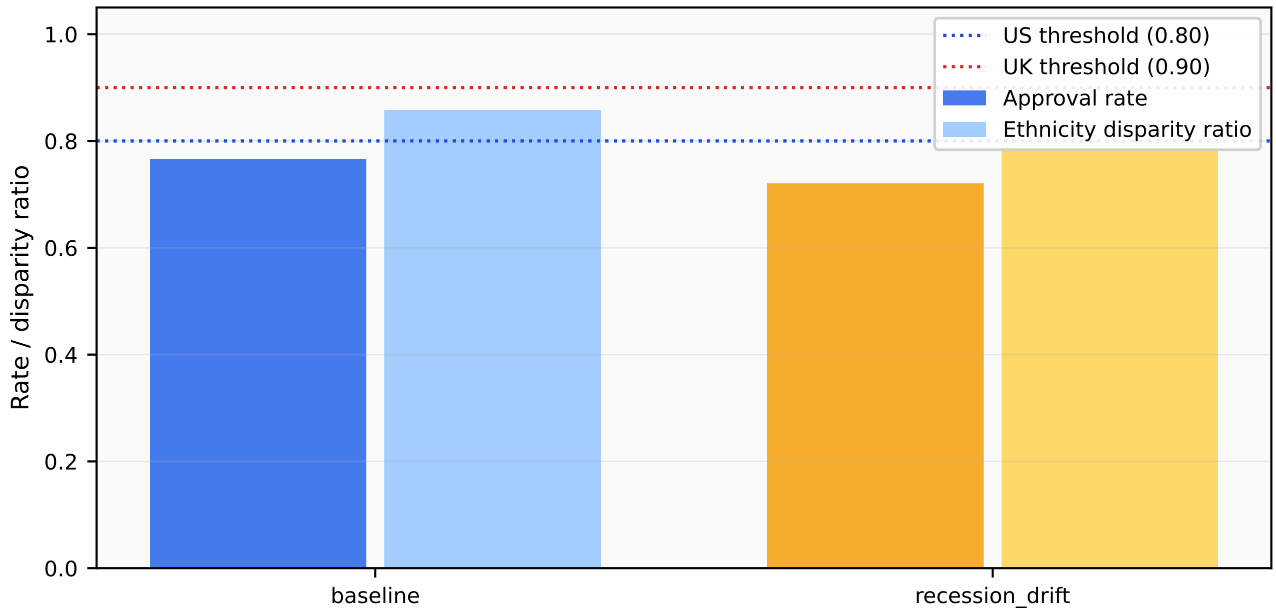

Figure 2: Drift simulation: approval rate and ethnicity disparity ratio under baseline and recession-scenario conditions. The baseline-trained model is applied unchanged to the income-shocked population. Dark bars show approval rate; light bars show ethnicity disparity ratio. The ethnicity disparity ratio falls from approximately 0.858 (baseline) to 0.799 (recession), crossing below the US threshold of 0.80 (blue dotted line). A portfolio that clears the current US compliance check would fail that same check under recession conditions — demonstrating that one-time model validation is insufficient and continuous post-deployment monitoring is required.

6. What Does the Tool Not Do, and Why Were Those Boundaries Chosen?

The tool does not use real customer data. All applicant records are synthetically generated. This means the fairness conclusions are illustrative rather than dispositive: the synthetic data cannot reproduce the actual joint distribution of income, credit score, and demographic group that would appear in a real lender's portfolio. This boundary was chosen because the assignment is a class prototype, not a production system. Using real data would require data-sharing agreements, privacy safeguards, and regulatory permissions that are outside the scope of the assignment.

The prototype implements SHAP (SHapley Additive exPlanations) using LinearExplainer for the logistic regression model. The notebook demonstrates both a global summary plot (distribution of SHAP values across all applicants) and a local waterfall plot (per-applicant feature contributions for individual rejected applicants). SHAP provides model-consistent attributions that satisfy the spirit of the CFPB's adverse action guidance: the explanation reflects what the model actually computed, not a post-hoc approximation. LIME is not implemented; it would add model-agnostic local explanations for non-linear models and would be the next addition if the credit model were upgraded to gradient boosting.

The adverse action reason module (src/model_pipeline.py: adverse_action_reasons) generates human-readable rejection text using feature-threshold heuristics — for example, flagging a credit score below 640 or a debt-to-income ratio above 0.45. This is a deliberate simplification for the prototype: these heuristics approximate the model's reasoning but are not derived directly from the model's coefficients or SHAP values. In a production system, the SHAP waterfall values would drive adverse action text generation, ensuring the explanation and the model's decision are derived from the same underlying computation. This gap is acknowledged as a design limitation.

The tool does not produce a compliance certificate or a single pass/fail score. This boundary is deliberate and reflects the values audit: a single score would encourage clients to treat the tool as an audit shield rather than a genuine risk instrument. The output instead surfaces specific, rule-linked findings that require human interpretation.

The regulatory thresholds (0.80 for US, 0.90 for UK) are illustrative. The 0.80 threshold draws on the four-fifths rule from US employment law, which is sometimes applied analogously to credit lending disparate impact analysis. The 0.90 threshold is a design choice reflecting the Consumer Duty's stricter outcome orientation. Both should be validated by qualified legal counsel and compliance teams before deployment. With more time, the next additions would be: SHAP explainability, a real lender data ingestion pipeline, automated rule-change alerts triggered by regulatory update feeds, approval workflow integration for human review escalation, and time-series monitoring dashboards replacing the current static drift simulation.

7. Failure Mode Analysis

The prototype explicitly identifies five failure modes, consistent with the assignment requirement that failure modes be named.

- False positives: the tool incorrectly flags a compliant portfolio. Lenders may respond by tightening credit criteria to clear the threshold, inadvertently reducing lending access to the groups the tool was intended to protect.
- False negatives: the tool clears a non-compliant portfolio. The lender reduces monitoring and defers remediation; hidden disparity persists and affected consumer groups face systematic exclusion without regulatory response. This is the more serious failure in terms of consumer harm.
- Performance drift: economic conditions or applicant mix change post-deployment, altering approval rates and group disparity ratios without the lender's awareness. Drift monitoring (Figure 2) is implemented but requires continuous real data ingestion to be effective in production.
- Jurisdiction misconfiguration: the US rule set is applied to a UK-regulated product or vice versa. The UK Consumer Duty evidence requirements are more extensive than the US configuration; misconfiguration would leave those requirements unchecked while management believes the product is compliant.
- False sense of compliance: the most dangerous failure. The tool produces thorough-looking reports and rule-linked findings while genuine harmful outcomes persist. This failure requires the tool to be honest about the limits of its own coverage — which is commercially inconvenient. The prototype addresses this by explicitly documenting what it does not check and flagging which findings require human judgement.